
Cyber Forensics - 24CY611

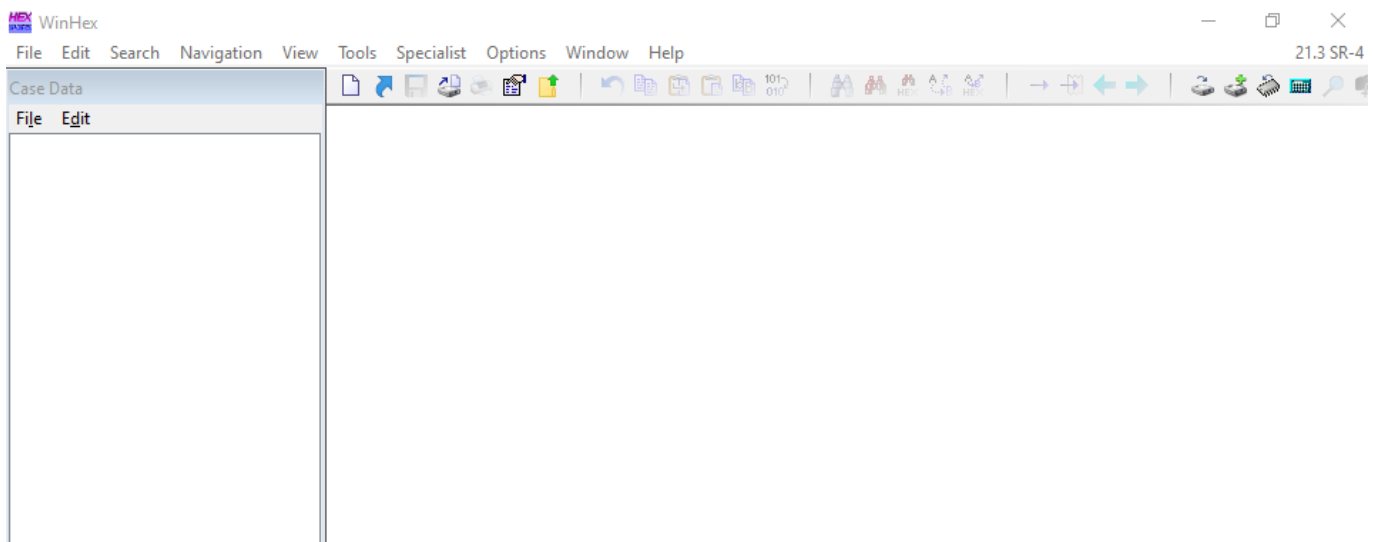
Lab 2 - Understanding Hard Disks and File Systems

1. Recovering Deleted Files from Hard Disks Using WinHex

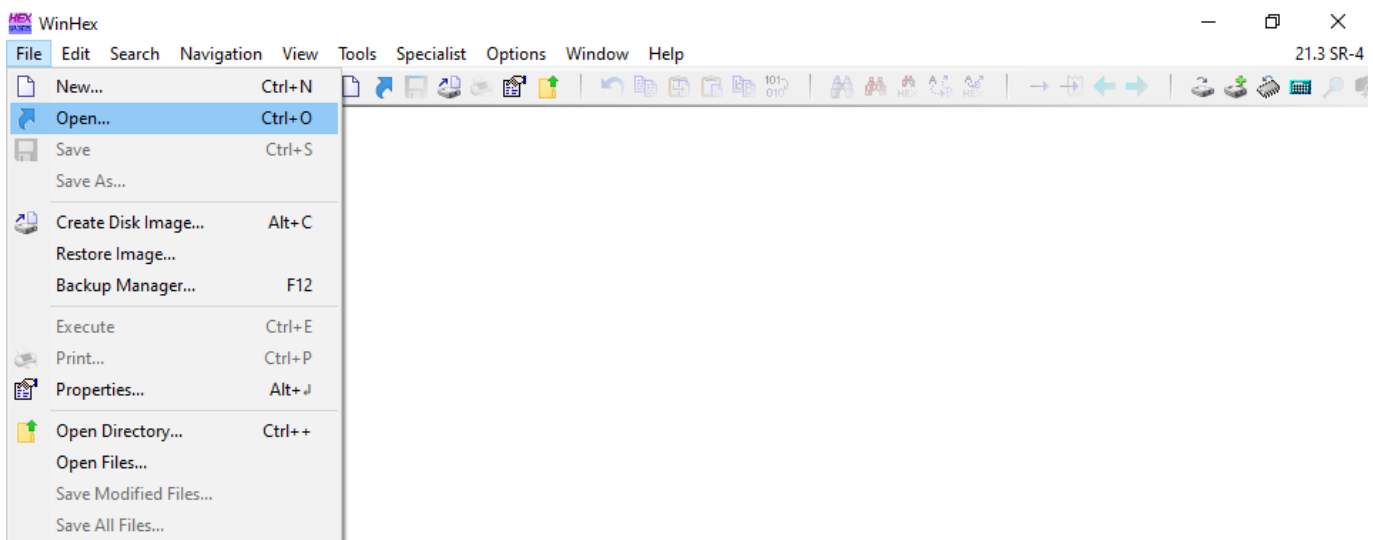
WinHex

WinHex is a versatile hexadecimal editor used for data recovery, digital forensics, and low-level file or disk analysis. It allows users to inspect and edit raw data on storage devices, memory, or files in hex format.

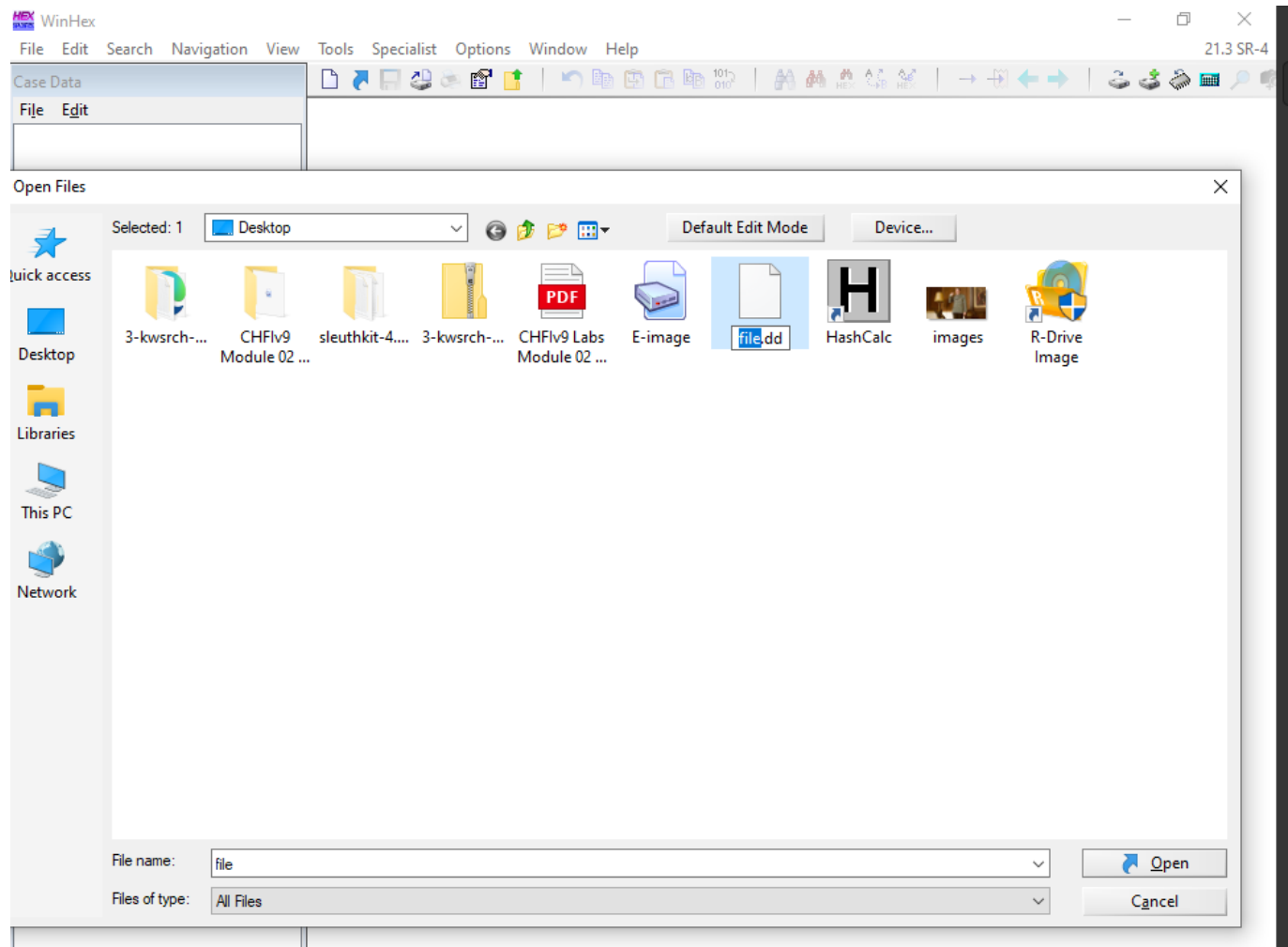
1. Install and open WinHex



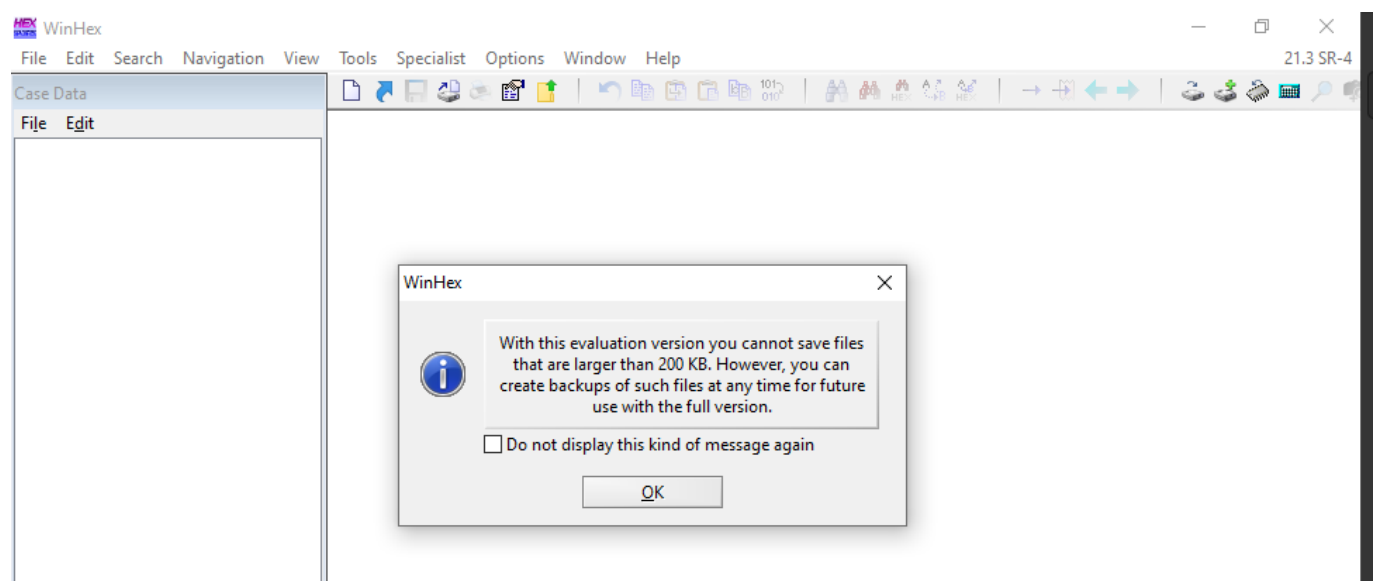
2. Navigate to File and click Open to add the evidence file



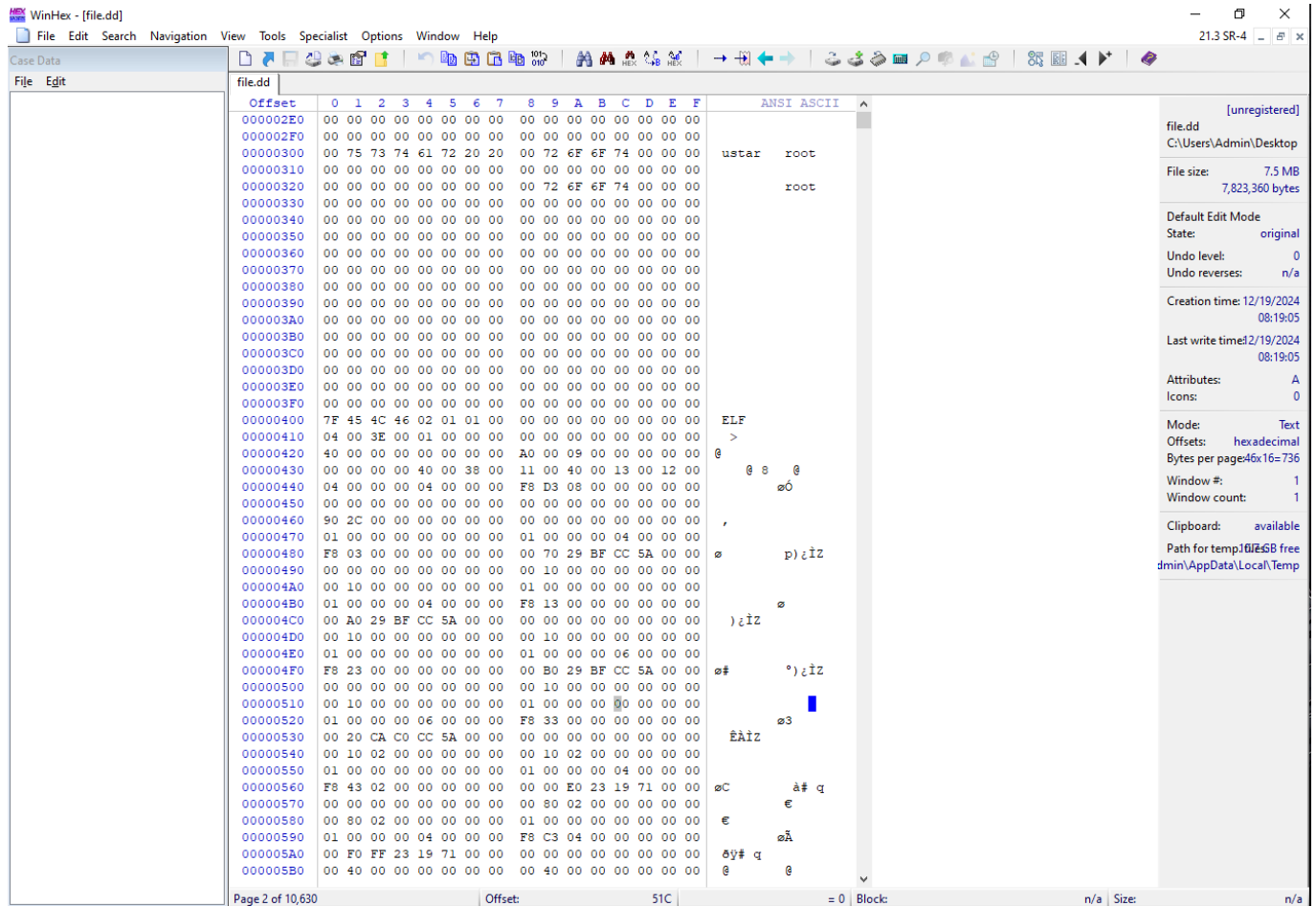
3. Then select the disk image file from the file explorer. Next click open .



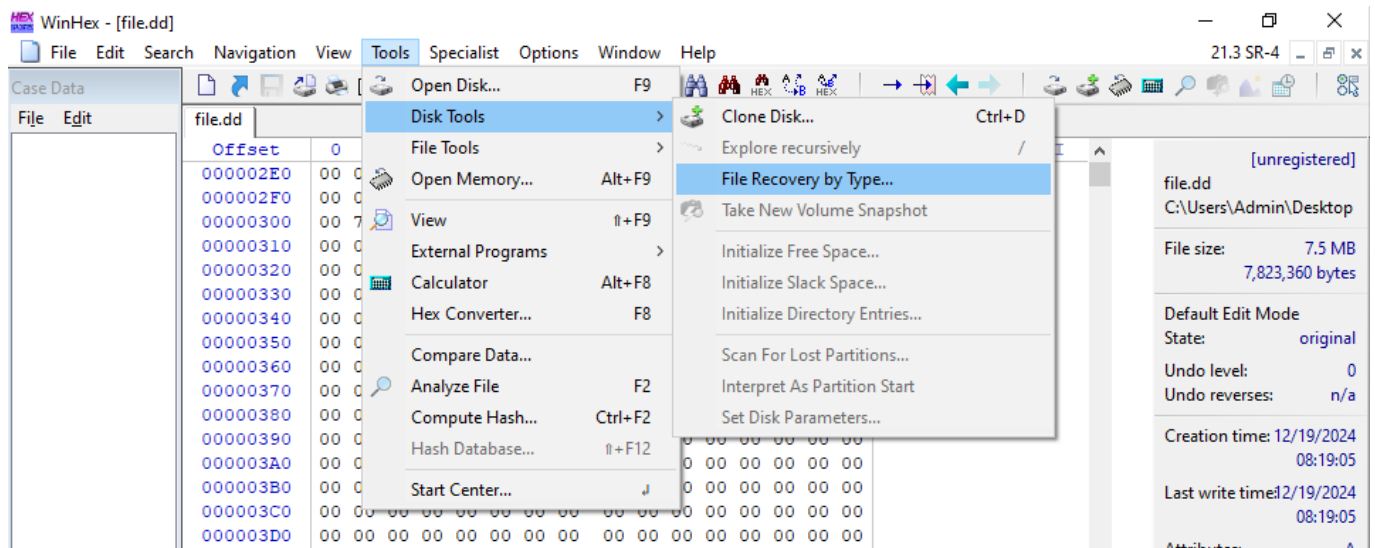
4. Then WinHex evaluation popup appears , click OK



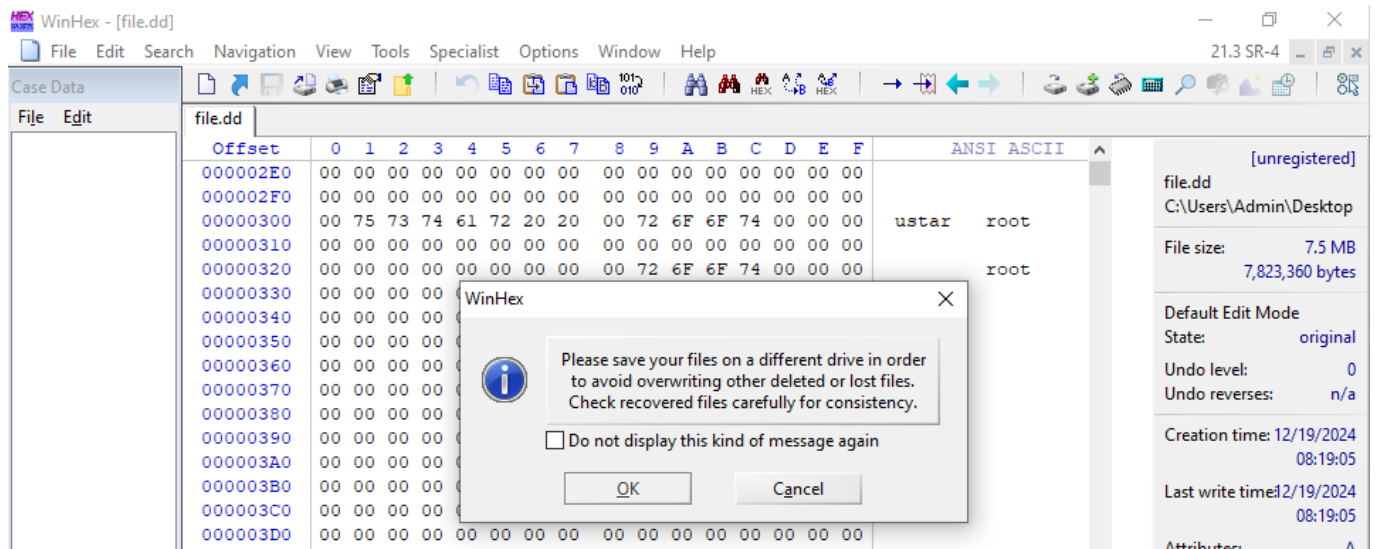
5. WinHex will process the image file and display the data as hexadecimal values in a Data Interpreter.



6. Navigate to Tools → Disk Tools → File Recovery by Type

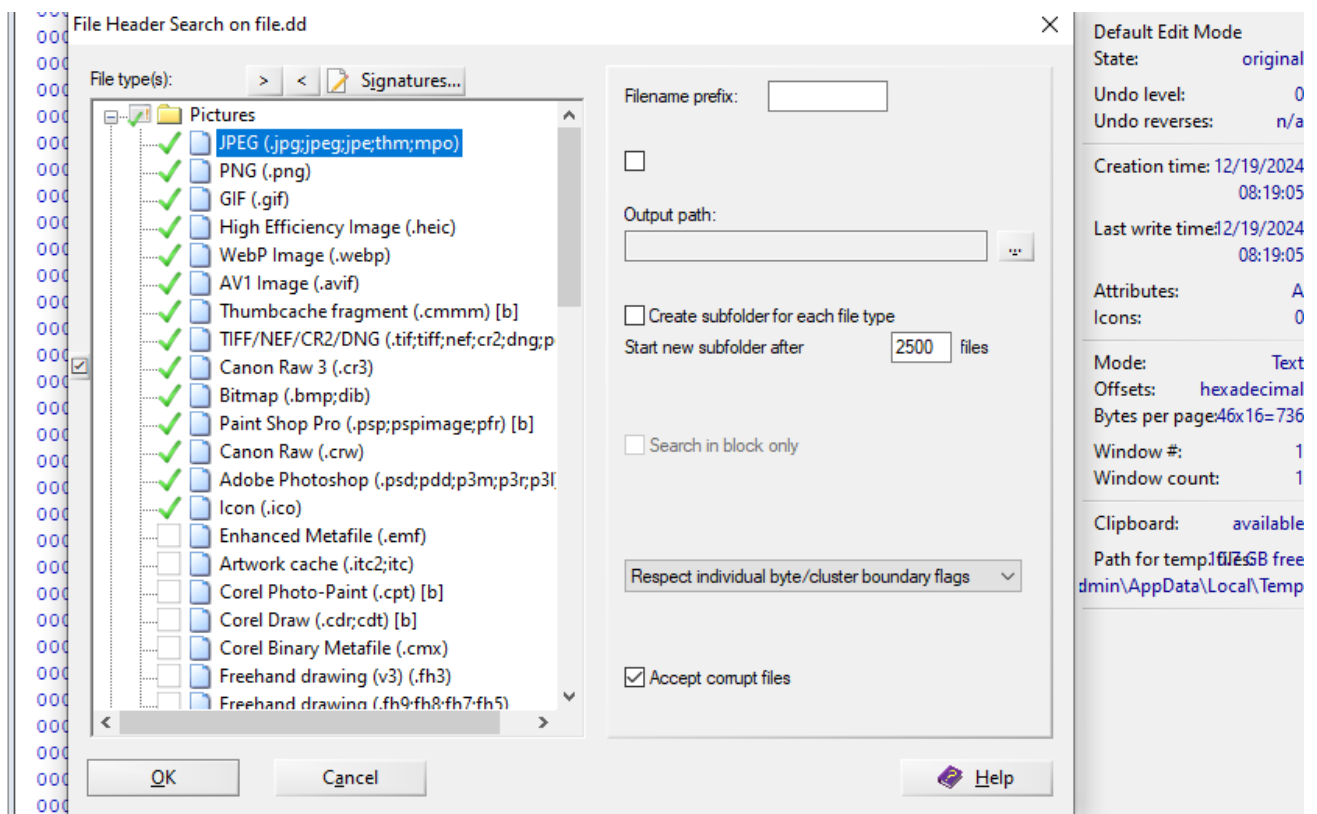


7. Click OK in the popup

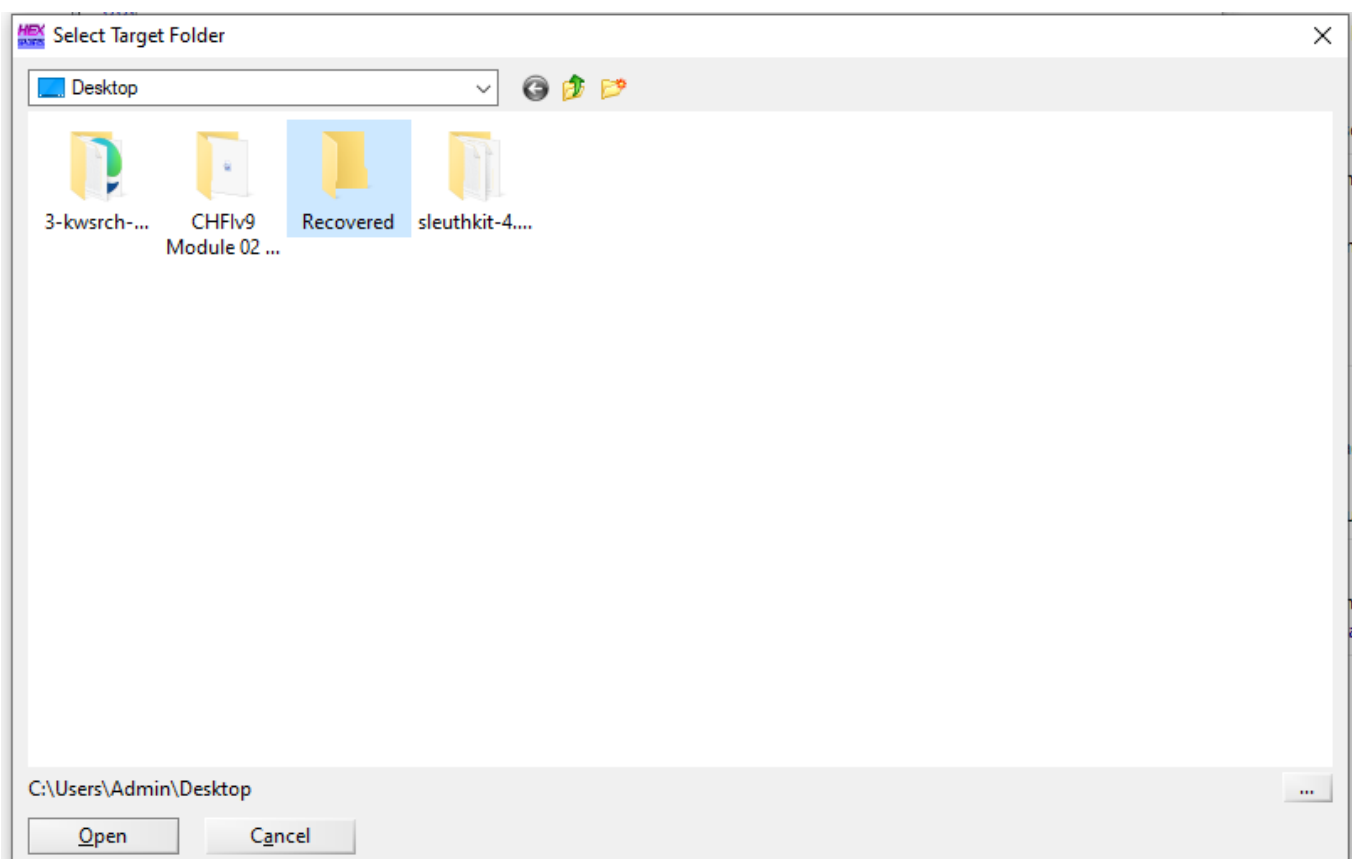


8. File header search on file.dd window appears. In this lab we are going to extract the pictures folder, click on + node to expand the pictures folder.

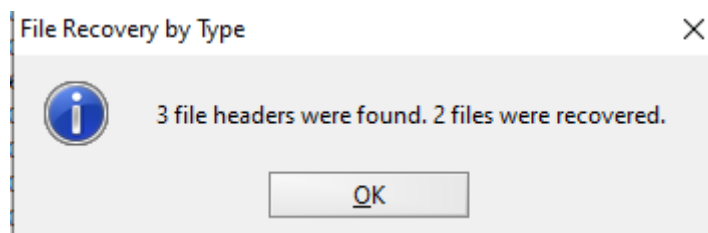
9. Select the file types of the target files to recover and then click OK.



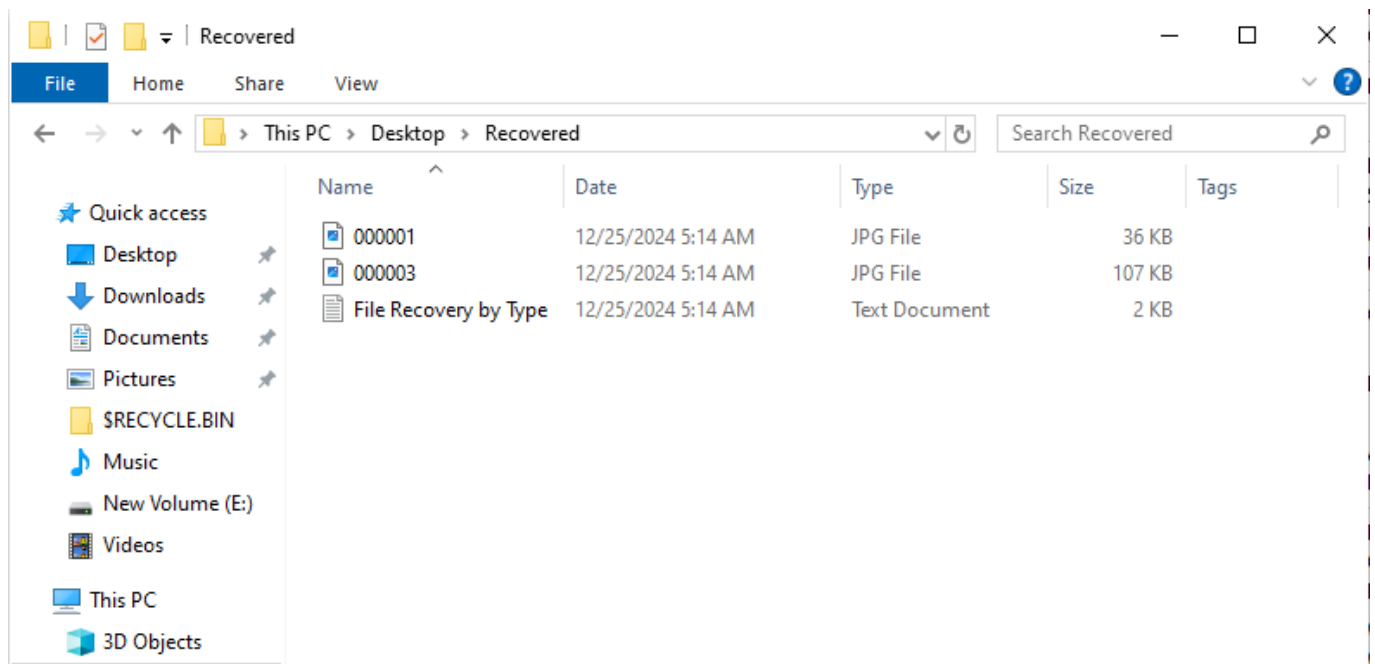
10. Select the target folder to save the recovered files and then click OK.



11. After recovery process is complete , click OK in the popup.



12. Open the target folder to see the recovered files.

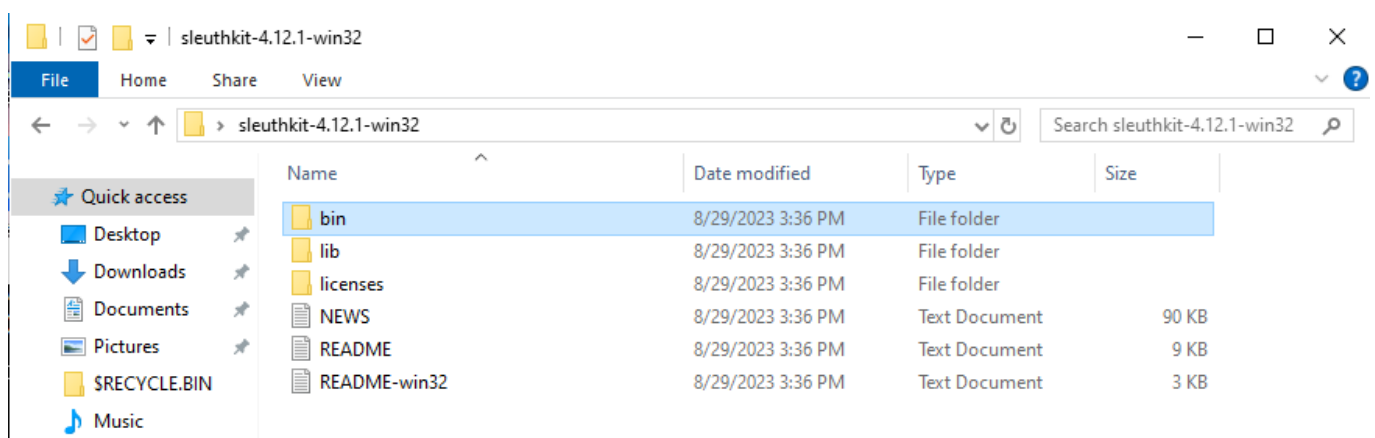


2. Analyzing File System Types Using The Sleuth Kit (TSK)

The Sleuth Kit

The Sleuth Kit (TSK) is an open-source digital forensics toolkit for analyzing file systems and recovering data. It supports various file systems like NTFS, FAT, and ext, enabling investigators to examine partitions, metadata, and deleted files. TSK's command-line tools provide detailed insights into file system structures and activities for forensic analysis.

1. Navigate to **C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin** and open with command prompt.



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>
```

2. Now type **fsstat -f "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"** and then press Enter to see the file system details.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>fsstat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: BA50ED9250ED5623
OEM Name: NTFS
Volume Name: KW-SRCH-1
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 5355
First Cluster of MFT Mirror: 8032
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 39
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 16063
Total Sector Range: 0 - 16063

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)  Size: 48-72  Flags: Resident
$ATTRIBUTE_LIST (32)       Size: No Limit  Flags: Non-resident
$FILE_NAME (48)           Size: 68-578   Flags: Resident,Index
$OBJECT_ID (64)           Size: 0-256   Flags: Resident
$SECURITY_DESCRIPTOR (80)  Size: No Limit  Flags: Non-resident
$VOLUME_NAME (96)         Size: 2-256   Flags: Resident
$VOLUME_INFORMATION (112)  Size: 12-12   Flags: Resident
$DATA (128)               Size: No Limit  Flags:
$INDEX_ROOT (144)         Size: No Limit  Flags: Resident
$INDEX_ALLOCATION (160)    Size: No Limit  Flags: Non-resident
$BITMAP (176)             Size: No Limit  Flags: Non-resident
$REPARSE_POINT (192)      Size: 0-16384  Flags: Non-resident
$EA_INFORMATION (208)     Size: 8-8    Flags: Resident
$EA (224)                 Size: 0-65536  Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536  Flags: Non-resident
```

3. Use the **istat** tool to view the details of metadata structure.

4. Now type **fsstat -f "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 0** and then press Enter.

```

C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>istat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 0
MFT Entry Header Values:
Entry: 0          Sequence: 1
$LogFile Sequence Number: 1075381
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 256 (S-1-5-21-1757981266-484763869-1060284298-1003)
Created:      2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:      2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $MFT
Parent MFT Entry: 5      Sequence: 5
Allocated Size: 16384     Actual Size: 16384
Created:      2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:      2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)  Name: N/A      Resident   size: 72
Type: $FILE_NAME (48-3)              Name: N/A      Resident   size: 74
Type: $DATA (128-1)                  Name: N/A      Non-Resident size: 39936 init_size: 39936
5355 5356 5357 5358 5359 5360 5361 5362
5363 5364 5365 5366 5367 5368 5369 5370
5371 5372 5373 5374 5375 5376 5377 5378
5379 5380 5381 5382 5383 5384 5385 5386
5387 5388 5389 5390 5391 5392 5393 5394
5395 5396 5397 5398 5399 5400 5401 5402
5403 5404 5405 5406 5407 5408 5409 5410
5411 5412 5413 5414 5415 5416 5417 5418
5419 5420 5421 5422 5423 5424 5425 5426
5427 5428 5429 5430 5431 5432 5433 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
Type: $BITMAP (176-5)  Name: N/A      Non-Resident   size: 8 init_size: 8
5354

```

5.To view MFTMirr File Overview, type **fsstat -f "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 1** and then press Enter.

```

C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>istat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 1
MFT Entry Header Values:
Entry: 1          Sequence: 1
$LogFile Sequence Number: 1052784
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 256 (S-1-5-21-1757981266-484763869-1060284298-1003)
Created:      2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:      2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $MFTMirr
Parent MFT Entry: 5      Sequence: 5
Allocated Size: 4096     Actual Size: 4096
Created:      2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:      2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)  Name: N/A      Resident   size: 72
Type: $FILE_NAME (48-2)              Name: N/A      Resident   size: 82
Type: $DATA (128-1)                  Name: N/A      Non-Resident size: 4096 init_size: 4096
8032 8033 8034 8035 8036 8037 8038 8039

```


The Master File Table (MFT) is a critical component of the NTFS file system, storing metadata about every file and directory on a disk. Each record in the MFT contains details such as file names, timestamps, permissions, and physical location on the storage device

6. To view Boot File Overview, type **fsstat -f "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"** 7 and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>fsstat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 7
MFT Entry Header Values:
Entry: 7          Sequence: 7
$LogFile Sequence Number: 0
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 0 ()
Created: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $Boot
Parent MFT Entry: 5      Sequence: 5
Allocated Size: 8192     Actual Size: 8192
Created: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 48
Type: $FILE_NAME (48-2) Name: N/A Resident size: 76
Type: $SECURITY_DESCRIPTOR (80-3) Name: N/A Resident size: 116
Type: $DATA (128-1) Name: N/A Non-Resident size: 8192 init_size: 8192
0 1 2 3 4 5 6 7
8 9 10 11 12 13 14 15
```

7. To view File Volume Overview, type **fsstat -f "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 3** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>fsstat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 3
MFT Entry Header Values:
Entry: 3          Sequence: 3
$LogFile Sequence Number: 1087457
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 0 ()
Created:          2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:    2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:     2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:         2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $Volume
Parent MFT Entry: 5      Sequence: 5
Allocated Size: 0        Actual Size: 0
Created:          2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:    2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:     2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:         2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$OBJECT_ID Attribute Values:
Object Id: 8edb73da-63d1-4dcd-b2d9-7035350ef062

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 48
Type: $FILE_NAME (48-1) Name: N/A Resident size: 80
Type: $OBJECT_ID (64-6) Name: N/A Resident size: 16
Type: $SECURITY_DESCRIPTOR (80-2) Name: N/A Resident size: 116
Type: $VOLUME_NAME (96-4) Name: N/A Resident size: 18
Type: $VOLUME_INFORMATION (112-5) Name: N/A Resident size: 12
Type: $DATA (128-3) Name: N/A Resident size: 0
```

8. To view AttrDef File Overview, type **fsstat -f "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 4** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>fsstat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 4
MFT Entry Header Values:
Entry: 4          Sequence: 4
$LogFile Sequence Number: 1053965
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 0 ()
Created:          2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:    2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:     2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:         2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $AttrDef
Parent MFT Entry: 5      Sequence: 5
Allocated Size: 36352    Actual Size: 36000
Created:          2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:    2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:     2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:         2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 48
Type: $FILE_NAME (48-2) Name: N/A Resident size: 82
Type: $SECURITY_DESCRIPTOR (80-3) Name: N/A Resident size: 116
Type: $DATA (128-4) Name: N/A Non-Resident size: 2560 init_size: 2560
5339 5340 5341 5342 5343
```

9. To view Bitmap File Overview, type **fsstat -f "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 6** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>istat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 6
MFT Entry Header Values:
Entry: 6          Sequence: 6
$LogFile Sequence Number: 1052934
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 256 (S-1-5-21-1757981266-484763869-1060284298-1003)
Created: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $Bitmap
Parent MFT Entry: 5      Sequence: 5
Allocated Size: 2048     Actual Size: 2008
Created: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-2) Name: N/A Resident size: 80
Type: $DATA (128-1) Name: N/A Non-Resident size: 2008 init_size: 2008
8128 8129 8130 8131

C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>
```

10. To view BiadClus File Overview, type **fsstat -f "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 8** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>istat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 8
MFT Entry Header Values:
Entry: 8          Sequence: 8
$LogFile Sequence Number: 1053084
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 256 (S-1-5-21-1757981266-484763869-1060284298-1003)
Created: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $BadClus
Parent MFT Entry: 5      Sequence: 5
Allocated Size: 0        Actual Size: 0
Created: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-3) Name: N/A Resident size: 82
Type: $DATA (128-2) Name: N/A Resident size: 0
Type: $DATA (128-1) Name: $Bad Non-Resident size: 8224768 init_size: 0

C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>
```

11. To view Secure File Overview, type **fsstat -f "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 9** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>fsstat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 9
MFT Entry Header Values:
Entry: 9          Sequence: 9
LogFile Sequence Number: 1086357
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 257 (S-1-5-21-1757981266-484763869-1060284298-1003)
Created: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed: 2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags:
Name: $Secure
Parent MFT Entry: 5      Sequence: 5
Allocated Size: 0        Actual Size: 0
Created: 2076-11-29 00:54:34.000000000 (Pacific Standard Time)
File Modified: 2076-11-29 00:54:34.000000000 (Pacific Standard Time)
MFT Modified: 2076-11-29 00:54:34.000000000 (Pacific Standard Time)
Accessed: 2076-11-29 00:54:34.000000000 (Pacific Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-7) Name: N/A Resident size: 80
Type: $DATA (128-8) Name: $SDS Non-Resident size: 264040 init_size: 264040
16 17 18 19 20 21 22 23
24 25 26 27 28 29 30 31
32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47
48 49 50 51 52 53 54 55
56 57 58 59 60 61 62 63
64 65 66 67 68 69 70 71
72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87
88 89 90 91 92 93 94 95
96 97 98 99 100 101 102 103
104 105 106 107 108 109 110 111
112 113 114 115 116 117 118 119
120 121 122 123 124 125 126 127
128 129 130 131 132 133 134 135
136 137 138 139 140 141 142 143
144 145 146 147 148 149 150 151
152 153 154 155 156 157 158 159
160 161 162 163 164 165 166 167
168 169 170 171 172 173 174 175
```

12. To vlist files and directory, type **fls -f "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>fls -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"
r/r 4-128-4: $AttrDef
r/r 8-128-2: $BadClus
r/r 8-128-1: $BadClus:$Bad
r/r 6-128-1: $Bitmap
r/r 7-128-1: $Boot
d/d 11-144-4: $Extend
r/r 2-128-1: $LogFile
r/r 0-128-1: $MFT
r/r 1-128-1: $MFTMirr
r/r 9-128-8: $Secure:$SDS
r/r 9-144-11: $Secure:$SDH
r/r 9-144-14: $Secure:$SII
r/r 10-128-1: $UpCase
r/r 3-128-3: $Volume
d/r 38-128-4: dir-n-6:there
d/d 38-144-1: dir-n-6
d/r 30-128-3: dir-r-4:there
d/d 30-144-1: dir-r-4
r/r 33-128-3: file-n-1.dat
r/r 35-128-3: file-n-3.dat
r/r 36-128-3: file-n-4.dat
r/r 37-128-3: file-n-5.dat
r/r 37-128-5: file-n-5.dat:here
r/r 27-128-1: file-r-1.dat
r/r 29-128-1: file-r-3.dat
r/r 29-128-3: file-r-3.dat:here
d/d 31-144-1: System Volume Information
-/r * 34-128-1: file-r-2.dat
V/V 39: $OrphanFiles
```

13. To view only deleted entries, type **fls -d "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>fls -d "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"
-/r * 34-128-1: file-r-2.dat
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>
```

14. To view details of an image file, type **img_stat "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>img_stat "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"
IMAGE FILE INFORMATION
-----
Image Type: raw
Size in bytes: 8224768
Sector size: 512
```

3. Analysing Raw Image using Autopsy

Autopsy is a user-friendly, open-source digital forensics platform for investigating data breaches, recovering deleted files, and analyzing disk images.

1. Open terminal in kali type `sudo autopsy` to start autopsy.

```
(bharath@kali)-[~]
$ sudo autopsy
[sudo] password for bharath:

Autopsy Forensic Browser
http://www.sleuthkit.org/autopsy/
ver 2.24

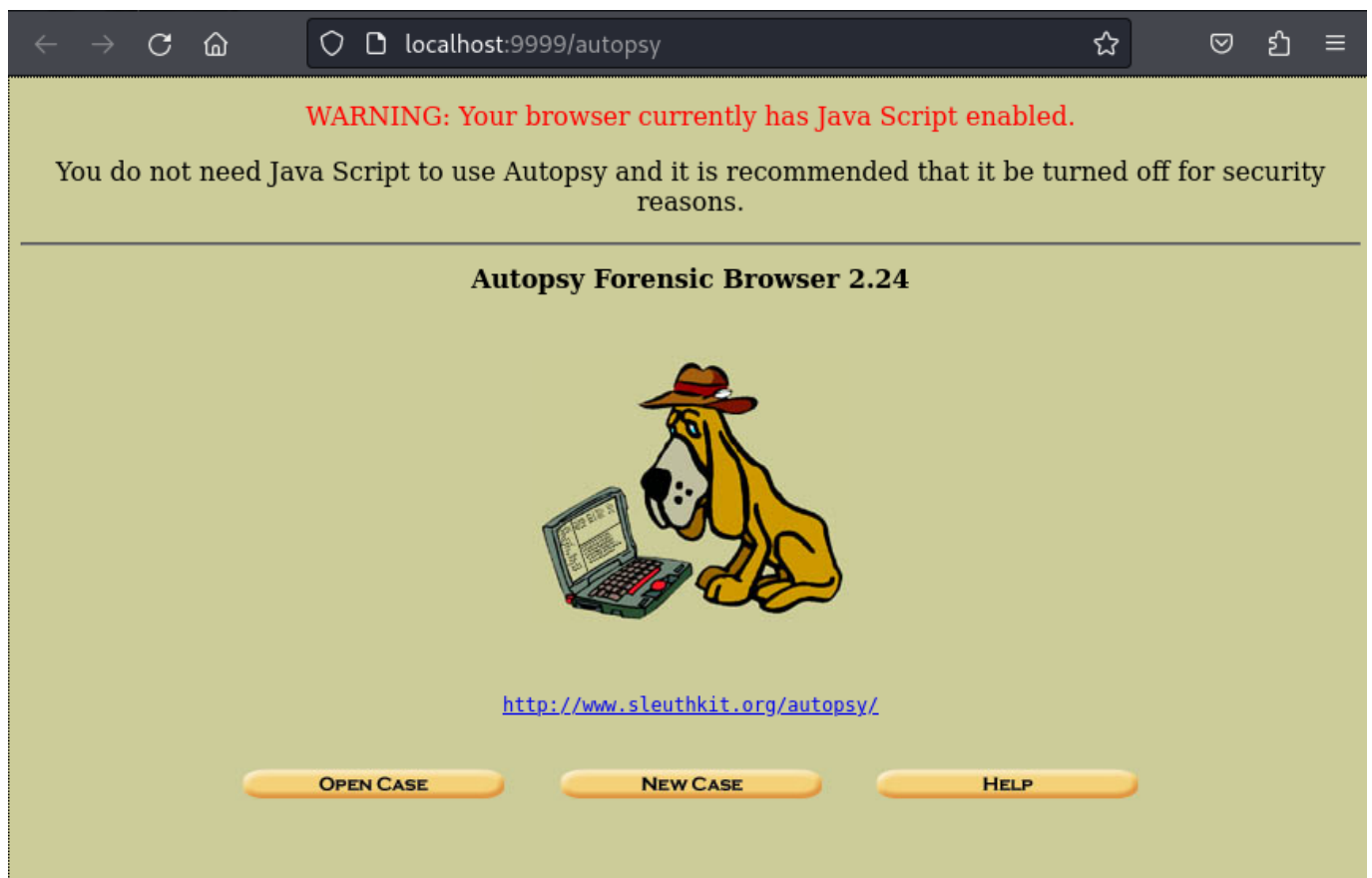
Evidence Locker: /var/lib/autopsy
Start Time: Wed Dec 25 09:05:10 2024
Remote Host: localhost
Local Port: 9999

Open an HTML browser on the remote host and paste this URL in it:

http://localhost:9999/autopsy

Keep this process running and use <ctrl-c> to exit
```

2. Then copy the link <http://localhost:9999/autopsy> in the terminal to browser. On loading the url autopsy main window will appear in the browser. Click **NEW CASE**.



3. Fill the required details in the **CREATE NEW CASE** page then click **NEW CASE**.

The screenshot shows a web browser window with the address bar displaying 'localhost:9999/autopsy?mod=0&view=1'. The page title is 'CREATE A NEW CASE'. The form is divided into three sections:

- 1. Case Name:** The name of this investigation. It can contain only letters, numbers, and symbols. The input field contains '100'.
- 2. Description:** An optional, one line description of this case. The input field contains 'Test case file'.
- 3. Investigator Names:** The optional names (with no spaces) of the investigators for this case. There are ten input fields labeled a. through j. Field 'a.' contains 'Bharath'.

At the bottom of the form, there are three buttons: 'NEW CASE', 'CANCEL', and 'HELP'.

4. Now click **ADD HOST** button

The screenshot shows a web browser window with the address bar displaying 'localhost:9999/autopsy?mod=0&view=2&case=100&desc=Test case file'. The page title is 'Creating Case: 100'. The page content includes:

Case directory (/var/lib/autopsy/100/) created
Configuration file (/var/lib/autopsy/100/case.aut) created

We must now create a host for this case.

Please select your name from the list: Bharath ▼

At the bottom, there is an 'ADD HOST' button.

5. Fill the details in the **ADD A NEW HOST** page and click **ADD HOST** button.

Case: 100

ADD A NEW HOST

- Host Name:** The name of the computer being investigated. It can contain only letters, numbers, and symbols.
- Description:** An optional one-line description or note about this computer.
- Time zone:** An optional timezone value (i.e. EST5EDT). If not given, it defaults to the local setting. A list of time zones can be found in the help files.
- Timeskew Adjustment:** An optional value to describe how many seconds this computer's clock was out of sync. For example, if the computer was 10 seconds fast, then enter -10 to compensate.
- Path of Alert Hash Database:** An optional hash database of known bad files.
- Path of Ignore Hash Database:** An optional hash database of known good files.

ADD HOST **CANCEL** **HELP**

6. Click **ADD IMAGE** button

Adding host: host1 to case 100

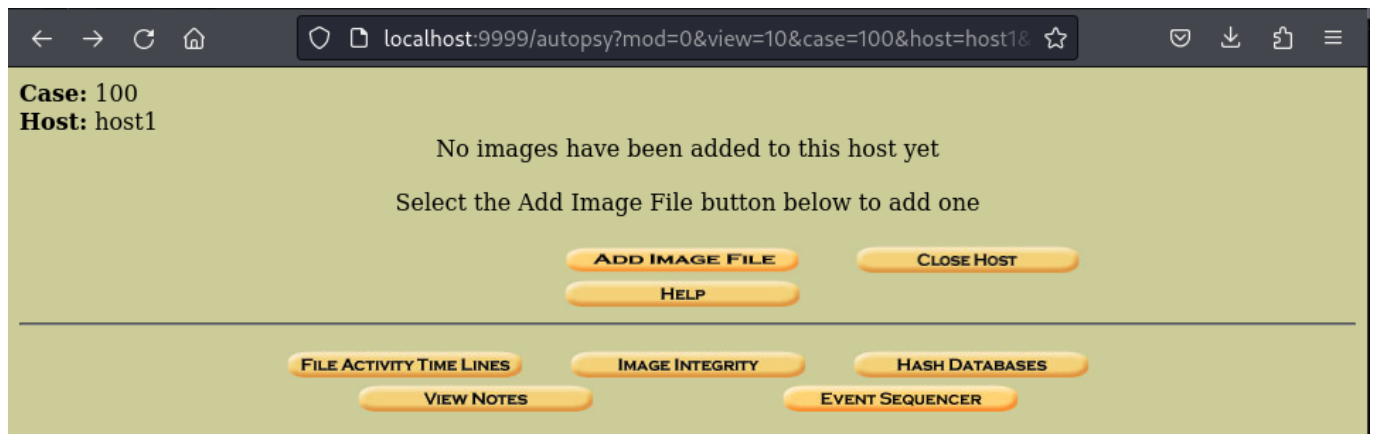
Host Directory (/var/lib/autopsy/100/host1/) created

Configuration file (/var/lib/autopsy/100/host1/host.aut) created

We must now import an image file for this host

ADD IMAGE

7. Click **ADD IMAGE FILE** button to add an image for investigation



8. Enter the image file location and select image type then click next.



9. Click **ADD** button in the Image file details page.

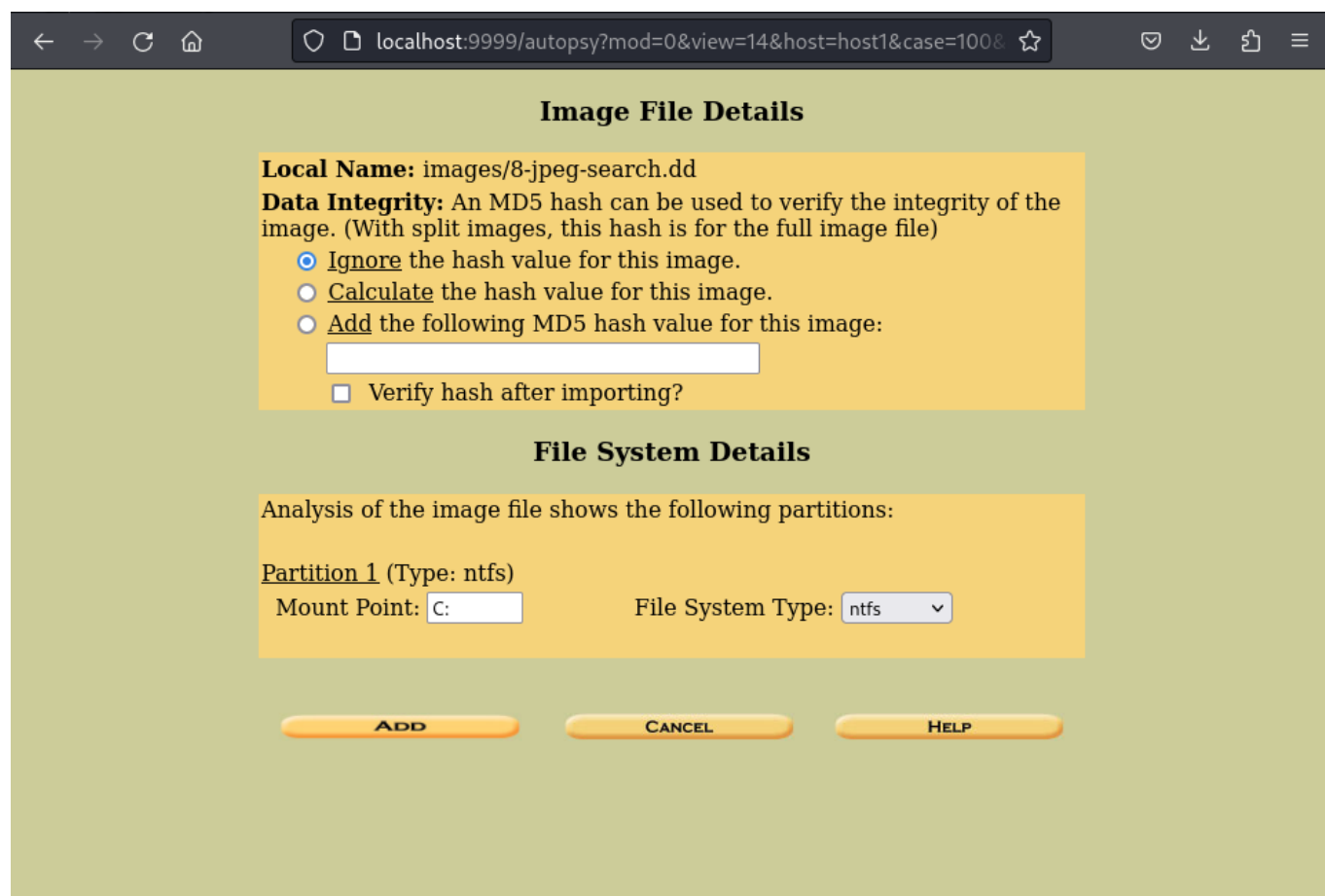


Image File Details

Local Name: images/8-jpeg-search.dd

Data Integrity: An MD5 hash can be used to verify the integrity of the image. (With split images, this hash is for the full image file)

☒ Ignore the hash value for this image.

☐ Calculate the hash value for this image.

☐ Add the following MD5 hash value for this image:

☐ Verify hash after importing?

File System Details

Analysis of the image file shows the following partitions:

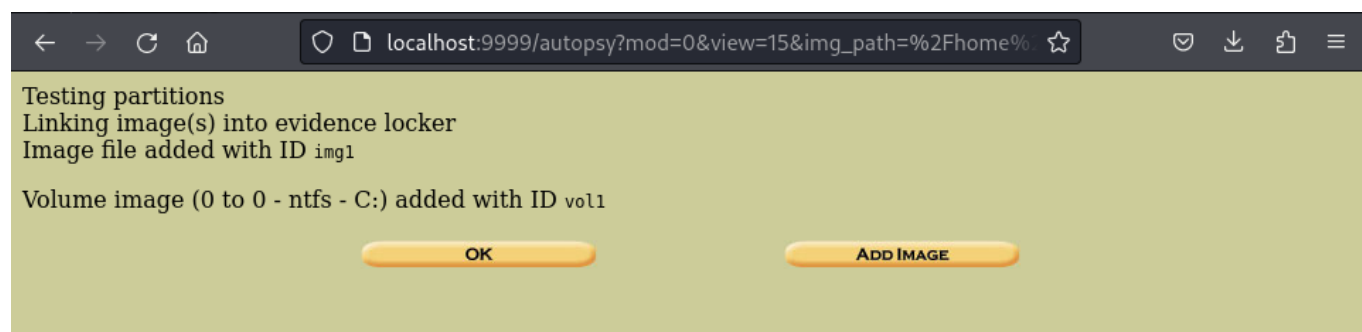
Partition 1 (Type: ntfs)

Mount Point:

File System Type:

ADD **CANCEL** **HELP**

10. Testing Partition page appears , click OK.



Testing partitions

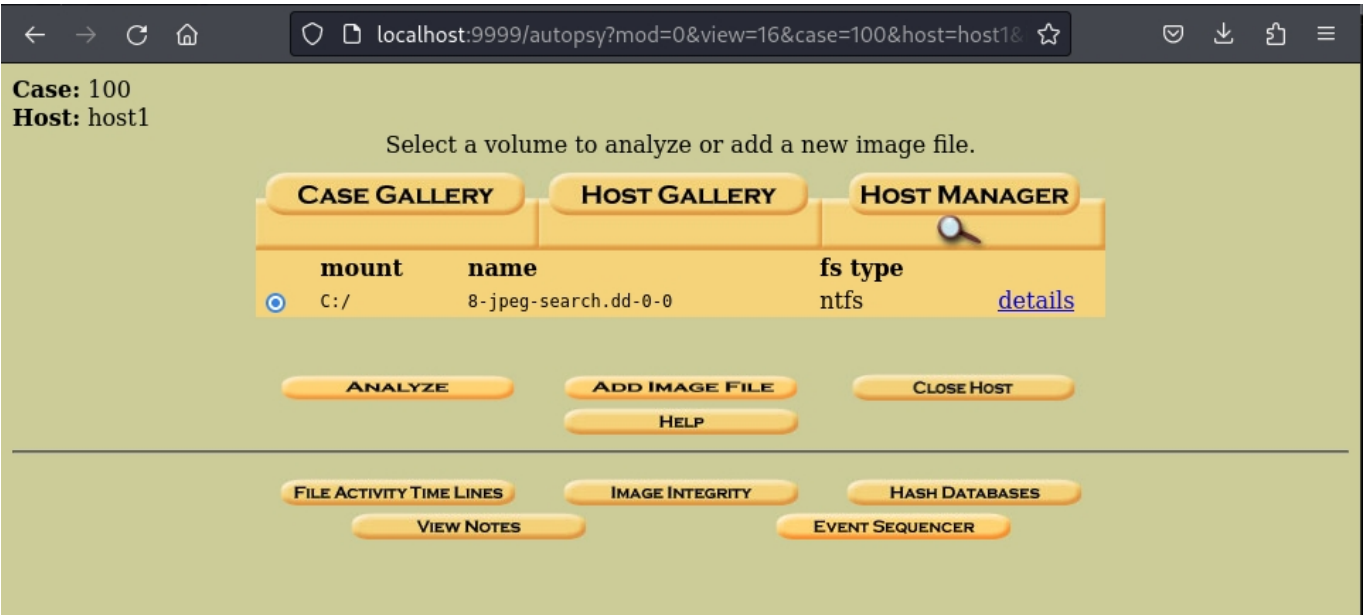
Linking image(s) into evidence locker

Image file added with ID `img1`

Volume image (0 to 0 - ntfs - C:) added with ID `vol1`

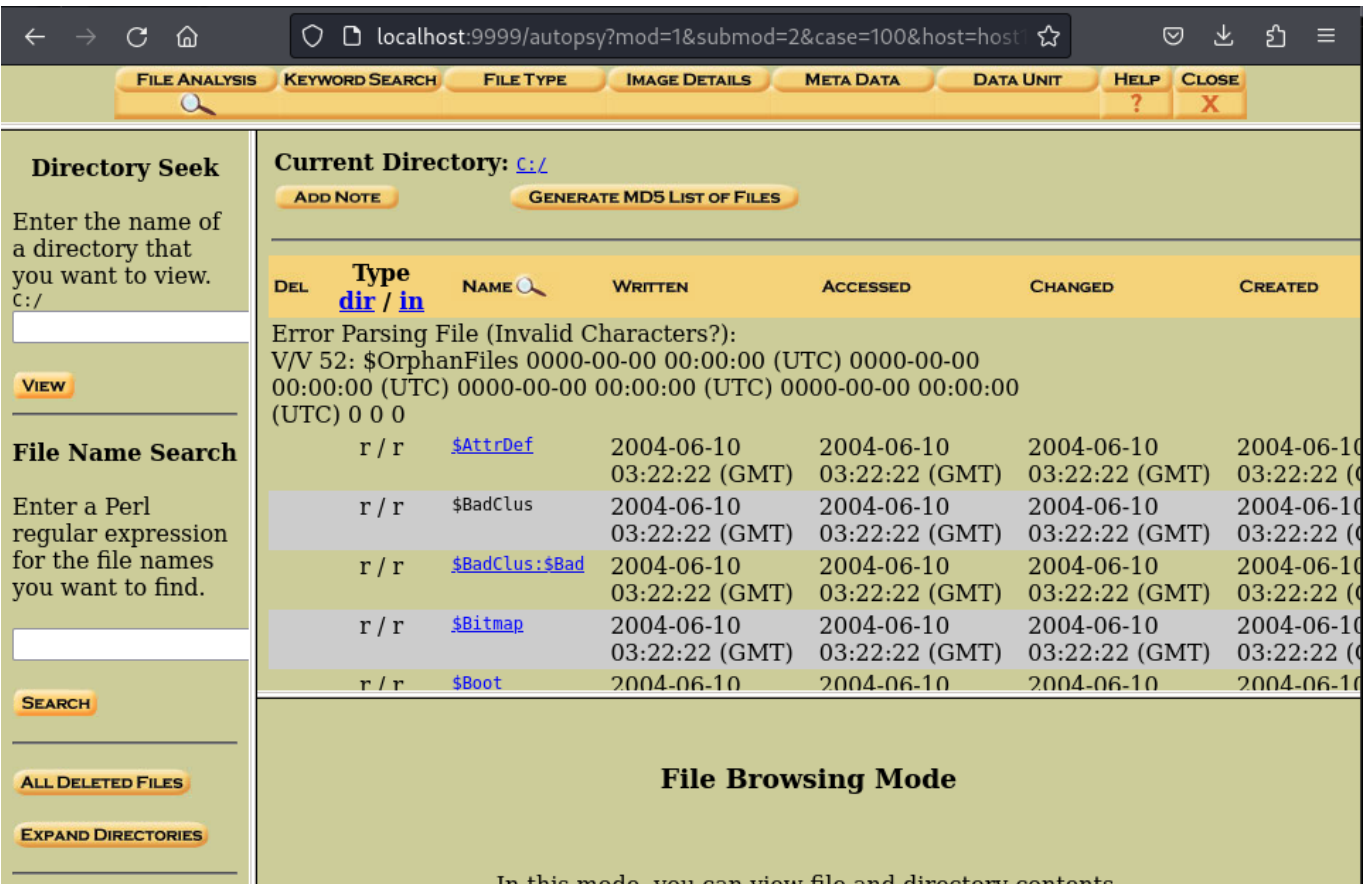
OK **ADD IMAGE**

11. Once the image added to the autopsy database, you can analyze the image. To analyze the image click **ANALYZE**

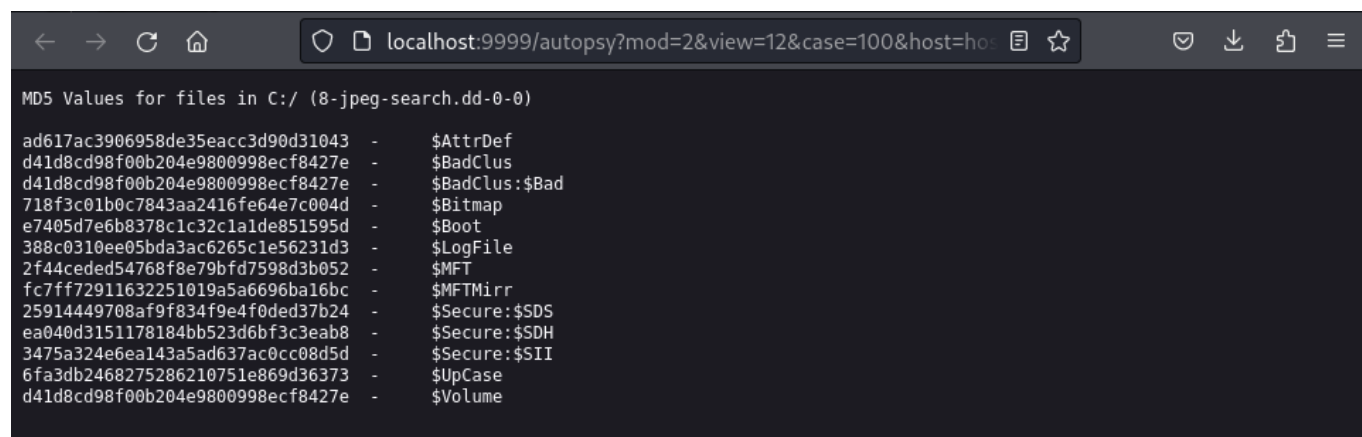


12. To do the file analysis, click **FILE ANALYSIS** button that allows you to analyze the image from the file and directory prespective.

13. File Analysis is used to examine the directories and files for evidence.



14. To generate MD5 hashes of the contained files , click **GENERATE MD5 LIST OF FILES** button.



```
MD5 Values for files in C:/ (8-jpeg-search.dd-0-0)

ad617ac3906958de35eacc3d90d31043 - $AttrDef
d41d8cd98f00b204e9800998ecf8427e - $BadClus
d41d8cd98f00b204e9800998ecf8427e - $BadClus:$Bad
718f3c01b0c7843aa2416fe64e7c004d - $Bitmap
e7405d7e6b8378c1c32c1a1de851595d - $Boot
388c0310ee05bda3ac6265c1e56231d3 - $LogFile
2f44ceded54768f8e79bfd7598d3b052 - $MFT
fc7ff72911632251019a5a6696ba16bc - $MFTMirr
25914449708af9f834f9e4f0ded37b24 - $Secure:$SDS
ea040d3151178184bb523d6bf3c3eab8 - $Secure:$SDH
3475a324e6ea143a5ad637ac0cc08d5d - $Secure:$SII
6fa3db2468275286210751e869d36373 - $UpCase
d41d8cd98f00b204e9800998ecf8427e - $Volume
```